

SHELL GLOBAL SOLUTIONS B.V.
LNG Terminal Expansion — Maasvlakte II
REQUEST FOR QUOTATION

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Project	LNG Terminal Expansion — Maasvlakte II, Rotterdam
Client	Shell Global Solutions B.V.
Contract Ref	SH-2024-LNG-EPC-001
Commodity	On/Off Gate Valves, Globe Valves, Check Valves — Cryogenic Service
Quantity	142 valves (see attached MTO)
Bid Deadline	2024-12-10 17:00 CET
Delivery Required	2025-06-30 (critical path item)
Incoterms	DAP Rotterdam (Maasvlakte II)
Currency	EUR

1. SCOPE OF SUPPLY

1.1 This RFQ covers the design, manufacture, inspection, testing, and delivery of cryogenic valves for the Shell Maasvlakte II LNG import terminal expansion. All valves shall be suitable for continuous operation in liquefied natural gas (LNG) service at temperatures down to -196°C.

1.2 The scope includes the following valve types:

Tag Range	Type	Size	Rating	Qty
10-GV-001 to 040	Gate Valve (slab, full bore)	2" – 24"	ASME Class 300	40
10-GV-041 to 080	Gate Valve (slab, full bore)	26" – 48"	ASME Class 150	40
10-GLV-001 to 022	Globe Valve (angle pattern)	1" – 6"	ASME Class 600	22
10-CV-001 to 040	Check Valve (tilting disc)	4" – 36"	ASME Class 150/300	40

2. MATERIAL REQUIREMENTS

2.1 Body & Bonnet

All valve bodies and bonnets shall be manufactured from ASTM A351 Grade CF8M (316L equivalent) austenitic stainless steel. Alternative: ASTM A182 Grade F316L for forged construction. Carbon steel bodies are NOT acceptable for any service in this specification.

2.2 Trim Materials

Stems shall be ASTM A276 Type 316L. Seats and discs shall be Stellite 6 hardfaced on 316L SS substrate. For gate valves in sizes $\geq 12"$, integral Stellite seat rings are mandatory — weld overlay is not acceptable.

2.3 Bolting

All pressure-retaining bolting shall be ASTM A193 Grade B8M (Class 2) studs with ASTM A194 Grade 8M heavy hex nuts. Low-temperature bolting per ASME B31.3 Table A-2 is mandatory. Carbon steel bolting (Grade B7/2H) is expressly prohibited.

2.4 Sealing / Packing

Valve packing shall be flexible graphite (Grafoil or equal) rated to -196°C. PTFE packing is acceptable only for globe valves ≤ 2 " in utility service. Asbestos-containing materials are strictly prohibited.

3. APPLICABLE STANDARDS & CODES

3.1 All valves shall be designed, manufactured, and tested in strict accordance with the following standards (latest edition unless noted):

Standard	Title	Applicability
ASME B16.34	Valves — Flanged, Threaded and Welding End	All valves — mandatory
API 598	Valve Inspection and Testing	All valves — mandatory
API 600	Bolted Bonnet Steel Gate Valves	Gate valves ≥ 2 "
API 623	Steel Globe Valves	Globe valves
ASME B16.5	Pipe Flanges and Flanged Fittings	Flange dimensions
BS 6364	Valves for Cryogenic Service	Mandatory for all LNG service
Shell DEP 80.30.00.19-Gen	Valves for Cryogenic and Low Temperature	Supersedes where more stringent
NACE MR0103	Materials Resistant to Sulfide Stress Cracking	Applicable if H ₂ S >50 ppmv
PED 2014/68/EU	Pressure Equipment Directive	CE marking mandatory
ISO 9001:2015	Quality Management Systems	QMS certification required

3.2 In case of conflict between standards, the more stringent requirement shall apply. Shell DEP takes precedence over API/ASME where more restrictive.

4. INSPECTION & TESTING REQUIREMENTS

4.1 Mandatory Testing

Test	Standard	Acceptance	Witness
Shell hydrostatic test (1.5× MAOP)	API 598 / ASME B16.34	Zero leakage	Shell TPA mandatory
Seat leakage test — liquid nitrogen	BS 6364 §8.3	Class A (zero leakage)	Shell TPA mandatory
Cryogenic function test (-196°C)	BS 6364 §8.4	Full stroke, zero binding	Shell TPA mandatory
NDE — Radiography (RT) of castings	ASME B16.34 App. A	Level 2 acceptance	Cert review
NDE — Liquid Penetrant (LPT) of welds	ASME V / VIII	Linear indications = reject	Cert review
PMI (Positive Material Identification)	Shell DEP	100% of wetted parts	Shell TPA mandatory
Hardness testing of trim	NACE MR0103	≤ 22 HRC	Cert review
Fugitive emissions test	ISO 15848-1	Class AH, Tightness B	Shell TPA mandatory

4.2 All test certificates shall be EN 10204 Type 3.2 (dual-certified by manufacturer and Shell-appointed TPA). Type 3.1 certificates are NOT acceptable for pressure-retaining parts.

4.3 Shell reserves the right to conduct unannounced shop visits at any stage of manufacture. 14 days written notice is required prior to any witness test.

5. COMMERCIAL REQUIREMENTS

5.1 Payment Terms: Net 60 days from invoice date. Payment milestone: 30% on order placement, 60% on FAT completion, 10% on delivery.

5.2 Warranty: 24 months from date of commissioning or 30 months from date of delivery, whichever is earlier. Extended warranty to 36 months is preferred and will be evaluated in bid scoring.

5.3 Liquidated Damages: 0.5% of order value per week of delay, capped at 10% of total contract value.

5.4 Spare Parts: Quotation shall include a recommended 2-year operational spares list. Commissioning spares to be included in base scope at no extra cost.

5.5 Lead Time: Maximum acceptable lead time is 22 weeks from purchase order to delivery Rotterdam. Bids exceeding 26 weeks will be disqualified.

6. DOCUMENTATION REQUIREMENTS

6.1 The following documentation shall be submitted with the bid:

- Technical data sheets for each valve type
- Compliance matrix against this specification (clause-by-clause)
- List of deviations and exceptions (mandatory — nil return if none)
- Quality plan / ITP (Inspection and Test Plan)
- CV and certifications of proposed Responsible Welding Coordinator
- Sample EN 10204 3.2 material certificates for reference order
- Current ISO 9001:2015 certificate
- List of three reference projects for LNG cryogenic valves (last 5 years)

6.2 Post-order documentation shall be submitted in English in PDF format via Shell's supplier portal (SIIS). Native CAD files (DWG/STEP) are required for custom items.

7. SPECIAL TECHNICAL NOTES

7.1 Fire-safe design per API 607 is required for all gate and globe valves. Note: Section 2.4 permits PTFE packing for small bore globe valves — vendors must confirm PTFE packing is acceptable for fire-safe certification or propose alternative.

7.2 All valve actuators shall be pneumatic (spring-return, fail-closed). Electric actuators may be proposed as an alternative for sizes ≤4" where pneumatic supply is unavailable.

7.3 Flange facing: RF (Raised Face) per ASME B16.5 for Class 150/300. Note: cryogenic service per BS 6364 recommends RTJ facing for Class 300 and above — vendor to advise preferred facing for LNG service.

7.4 Maximum allowable packing emission per ISO 15848-1 is Class AH. However, Section 4.1 test requirement references Class BH acceptance — vendor to clarify which class applies and confirm test capability.

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